Status	Finished
Started	Monday, 19 October 2026, 12:11 PM
Completed	Monday, 19 October 2026, 12:21 PM
Duration	9 mins 59 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(1)
b. O👎 ✓
c. O(n log n)
d. O(log n)

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is unsorted
b. When the list is very small
c. When the list contains duplicate elements
d. When the list is sorted ✓

Question **3**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the right sublist
b. The search ends successfully ✓
c. The search continues in the left sublist
d. The list is sorted

Question **4**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- a. The search is considered unsuccessful ✓
- b. The first element is returned
- c. An error is raised
- d. The last element is returned

Question **5**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

- a. By using a hash function
- b. By comparing each element sequentially
- c. By dividing the list length by two ✓
- d. By starting from the first element

Question **6**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Both (1) & (3)
- b. Binary ✓
- c. Linear
- d. Hash

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Linear search ✓
- b. Dynamic search
- c. Binary search
- d. Hashing

Question **8**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and unsorted
b. When the list is small and unsorted ✓
c. When the list is large and sorted
d. When the list is small and sorted

Question **9**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. 1
b. n ✓
c. log n
d. n/2

Question **10**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 94
b. 89 and 99
c. 90 and 94
d. 90 and 99 ✓

Question **11**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. log n
b. n ✓
c. n/2
d. n-1

Question **12**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Deleting elements from a list
b. Determining whether an element is present in a list ✓
c. Sorting elements in a list
d. Inserting elements into a list

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. There must be a mechanism to access middle element directly
b. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
c. Binary search algorithm is not efficient when the data elements more than 1500 ✓
d. Must use a sorted array

Question **14**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. At the beginning of the list
b. In the left sublist
c. In the middle of the list
d. In the right sublist ✓

Question **15**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Binary search

b. Both (1) & (3)

c. Hash search

d. Linear search ✓